Who enforces the laws of physics? No one, if matter is the relation itself

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Abstract

Philosophers ask who or what makes the laws of physics hold everywhere and always. This deposit, written for philosophers of physics, sets out the author's answer: matter does not obey relations, it is the relations, so the laws need no guardian, and their necessity is a necessity of identity, as in arithmetic. It shows the thesis at work on a clock, where two declared premises give Einstein's proper time as the geometric mean of two coprime light sides, and it shows how the same framework reads the dimensions of space, objects, mass, quantum probability and the direction of time. It also says which sense of «non-local», the foundation of the framework, is meant: not located in space and time, with non-separability as a consequence; Bell correlations are read with a retrocausal loop with local mediation, and action at a distance is excluded. For each thesis it says what rests on registered calculations, what is a premise and what is open.

Mathematical Materiality for philosophers: the Great Reversal, how it is carried out on a clock, how it reads space, objects, mass, quanta and time, and the author's answers to four objections.

Ugo Lanciano · ORCID 0009-0008-7618-9635 · Mathematical Materiality (MM) 2.68.1 · 26 October 2026

Version 2.68.1, 26 October 2026: section 6 reports a single calculation that puts the loop, the record and the direction of time into one model, with the author's answer on what the future chooses; the first open question of section 10 is closed. Everything else is the text of 2.68 (DOI 10.5281/zenodo.23669890); the materials of the calculation are in the folder CONTO_UNICO.

The Italian text is in LEGGIMI_IT.md. The author's sentences are quoted here in English; the Italian originals, with dates and context, are in AUTHOR_BACKGROUND.md.

The question

The laws of physics hold everywhere and always, and no one enforces them: what guarantees them? Mathematical Materiality (MM) answers that matter does not obey a relation: it is the relation. This text states the thesis, carries it out on a clock with two declared premises, and shows how the framework reads space, objects, mass, quanta and time.

In three lines. The laws have no guardian: a thing that broke them would not be disobedient, it would be another thing. There is no world behind the relations: the world we see is the web of relations read from inside. Mathematics works on the world because it is the form of a relation of the world, freed in the mind.

The two classical answers leave a gap between the rule and matter. If the law lives in a heaven of forms, it is unclear how it touches matter; if it lives in our language, it is unclear how it guesses the world. The simulation hypothesis puts a programmer in place of the heaven, and Vazza (2025) estimated that no one could pay for the energy.

The **Great Reversal** does not bridge the gap: it says there is no gap. Science has already made this move five times: caloric, phlogiston, vital force, the ether and the prototype kilogram in Paris looked like substances and turned out to be relations. Each time the world stayed the same; MM applies the same move to the last two items, matter and law.

Eight words of the framework. They appear throughout the text; their sense is enough here.

- **Nexus**: the starting link between two distinct terms, like 3 and 5.
- **Non-local**: not located in space and time; this is the pre-geometric sense, distinct from action at a distance (§1).
- **GCD**: the greatest common divisor, that is, the only part the two terms have in common; for 3 and 5 it is 1.
- **Copies**: the repetitions of the common unit when the nexus is counted.
- **Matrix**: the set of multiples and fractions of the GCD, read by whoever is inside it; the multiples appear as time, the fractions as space.

- **Diffraction**: the way the observer's matrix divides into fractions to give an object.
- **Record**: what an observer has collected; in the calculations it is the AI's translation of the observer's matrix.
- **The AI's translation**: the way a sentence of the author becomes a computable quantity. It is a choice, and the text marks it; an outcome that depends on a translation concerns that translation, not the premise.

The text also contains some **explanatory premises (AI)**. They are bridges proposed by the AI between a sentence of the author and its translation, written after an external review and checked by a second system (Gemini); the author accepted them into the text on 26 October. They are not theses of the author, and they leave him open questions (§10).

1. The nexus is a relation between two coprime numbers

The philosophical premises of MM fit in five sentences, and they have the logical form of the relation between 3 and 5. They are the postulates of MM, like Einstein's: they are not proved, they are judged by their consequences.

1. At the start there is not a thing but a difference: A is different from B.
2. To exist is reciprocal. Two entities exist for each other only through what they have in common; with nothing in common they could not even be called different.
3. Two entities given nothing but being two have no composite parts in common: they are **coprime**, like 3 and 5. A larger common part, like the 3 shared by 6 and 9, would be a third entity and would ask for one more reason.
4. Only the unit remains in common: GCD(3, 5) = 1. To belong to both, it has to be counted in each, three times in one and five in the other. Whatever something repeats in has parts, positions, distances: it has become local.
5. Yet three copies of the unit are the same unit counted three times, and between two copies of the same thing there is no distance. The local carries within itself, disguised as a number, the distance-free link it started from.

The first nexus is the one between space and time. In its non-local phase space and time are terms without measure: there is only a logical distinction, space is not time, and the two share only their GCD. Counted, they become the space and time we measure. This phase is not a distant level: it is present in every place and at every instant.

Four features of the relation between 3 and 5 are the features of the nexus:

- **No intermediary.** 3 and 5 are linked by the 1 with no number «in between» that carries the link. This is what «non-local» means: the relation is not located in a place or at an instant, and it has no path.
- **The terms do not come before the relation.** The 1 is not a brick waiting to be used: it appears as the only thing the two have in common.
- **Counting makes space and time.** The least common multiple, 15, contains the 3 five times and the 5 three times. In the matrix that arises from counting, the multiples appear as time and the fractions as space. With more counts (§2) time is their common growth, space their spread.
- **Counting preserves the nexus.** GCD × LCM = 1 × 15 = 3 × 5. However one counts in the local, the product remains that of the starting relation.

Explanatory premise (AI). The numbers are not quantities of space and time: in the non-local phase space and time are terms without measure. 3 and 5 are the arithmetic form of a logical situation: two distinct terms that have only a unit in common. Counting enters with premise 4: what is common, to belong to both, must appear in each, and appearing several times is already counting. That the two counts differ follows from premise 1: two equal terms would be one. A concrete case appears in the clock model (§2): there the step at rest (1, 1) does not distinguish the two light directions, while the asymmetry between two coprime counts, such as (3, 5), is what is read as motion.

On time and space the AI's reading is this. Multiplying and dividing are the two operations that can be performed on the common unit: multiplication gives an ordered succession of growing totals, division gives distinct parts of a total that stays the same. Read from inside, the first is what we call time and the second what we call space: time and space do not precede counting, they are the physical names of the accumulation and the partition of the same unit.

The author puts it this way (10 October): «there is an underlying step of reality where a non-local layer of material logic has in itself the features to generate the overlying step of space and objects». The common mistake, he adds, is to start from an objective space as a necessary element. In the volume MM 2.30 this is the pre-geometric programme; the sections that follow say how much of it has been carried out.

What «non-local» means

Non-locality is the foundation of the framework, and in the philosophy of physics the word has at least four senses. MM uses one, derives a second as a consequence and excludes the other two.

Sense	What it means	MM's position
Non-location (pre-geometry)	The relation is not located in a place or at an instant	This is MM's sense, confirmed by the author on 13 October
Non-separability (holism)	The whole does not reduce to parts separated in space, each with a state of its own	It follows: the copies of a nexus are the same unit, not separate parts
Bell non-locality (Bell 1964)	Correlations that no account with local properties, influences no faster than light and measurement settings independent of the state of the source reproduces	The measured correlations are a datum. Of the three assumptions, the premise drops the first at the root (separability: each system's own properties); the loop model with local mediation (§5) drops the third (independence of measurement settings). No influence at a distance
Action at a distance	An influence or a signal passes from one place to another faster than light	Excluded: no signals, and Tsirelson's bound is respected (§5)

The author's words. Besides the sentence of 10 October, three premises say how the non-local enters the local world. On 13 October, to the AI's reading that «non-local» means not located in space and time, the author answered: «yes, it holds for the reading too». On 24 September: «the original GCD is the passage from the non-local stage to the [local] stage of reality […] the matrix arises as a diffraction of the GCD in space and in time as separate elements». On 11 October: «it is the non-local basis of reality that forces the so-called present to a partial portion of the matrix to which the observer belongs; the non-local link "switches on" only part of the total matrix».

Explanatory premise (AI): nowhere, present everywhere. What is in no place is not elsewhere: it has no where. In the local it appears as a copy, and by premise 5 every copy is the same unit counted once more. This is why the non-local phase is present in every place and at every instant without being spread out in space: every place and every instant is a copy of it. Non-separability follows: two copies of the same unit are not two things with states of their own, but two readings of the same nexus.

Explanatory premise (AI): the passage and the present. In the author's sentences the original GCD is the hinge: where space and time, only distinct, find the common element, the nexus begins to be counted and the matrix arises, that is, the local world. The non-local phase does not remain a distant origin: according to the premise of 11 October it selects, within the matrix, the «switched-on» part that is the observer's present; and, by the answer of 13 October, the sense of «non-local» also holds for the reading. What makes the count begin, what distinguishes the switched-on part from the rest and in what sense the reading is non-local are open questions (§10).

What is not claimed. The non-local phase transmits no signals or influences, and today it has no distinct experimental signature: in the framework it is a premise. Its role is to give a reason for two facts: that the laws are the same everywhere (§7) and that the two wings of an entangled system are a single nexus (§5).

The neighbours. The idea that what appears separate is enfolded in an undivided whole is Bohm's implicate order (1980); the idea that the connectedness of space comes from entanglement is Van Raamsdonk's (2010). Non-separability as a thesis distinct from locality is Howard's (1985), who reads it in Einstein, and Teller's (1986), with relational holism. On the relation between Bell non-locality and relativity the reference is Maudlin (2011).

!Figure 1. One count gives time, space, mass and objects.

Figure 1 (figure_nexus_EN.svg). The multiples and fractions of the same GCD, read by whoever is inside the matrix, are time and space; contained or shared, they are mass and objects. Sections 2-6 follow this scheme from top to bottom.

2. The reversal in practice: a clock made of 3 and 5

The reversal is a procedure. On a clock, with two declared premises, it can be carried out down to the number: the clock that counts copies shows Einstein's proper time.

1. **Remove the substance.** Take a quantity that looks like a thing, flowing time, and ask which relation it is. In MM the time of a clock comes from the copies it counts.
2. **Write the relation as a pair.** An elementary step has two light sides, a and b. Premise (B in deposit MM 2.40): the two sides are coprime, and the step is indivisible; at rest it is (1, 1). Copies arise at the common multiples, so the light diamond holds $a \times b = \mathrm{GCD} \times \mathrm{LCM}$ cells.
3. **Combine the directions.** Another premise (C): two opposite directions are not added, they are multiplied. The clock shows $\sqrt{(a \times b)}$, the geometric mean; the laboratory, which adds, measures $(a + b)/2$, the arithmetic mean.
4. **Compare with the known law.** With 3 and 5 the clock shows $\sqrt{15} \approx 3.873$ and the laboratory 4: ratio 0.968.

The speed of the step is (5 − 3)/(5 + 3), a quarter of the speed of light, and relativity gives for that speed √(1 − 1/16) = 0.968.

Explanatory premise (AI). Why the product and not the sum: the fourth feature of the nexus says that counting preserves the product, and the copies of a step fill a × b = GCD × LCM cells. The duration of the step is therefore measured with an operation that preserves the product: the product ab, and its square root √(ab) to return to the scale of a single count, like the side of the square with the same number of cells as the a × b rectangle. The sum a + b instead changes when the asymmetry between the two sides changes.

This also shows which changes of observer preserve the nexus: those that lengthen one side in proportion as they shorten the other, a → ka and b → b/k, because they leave the product ab unchanged. Bondi's k factor thus enters as the transformation that preserves the product, not as an assumption placed before it.

Einstein's factor enters with premise C; from there on it is an identity. The arithmetic mean exceeds the geometric mean except when the sides are equal, and their ratio is always √(ab)/((a + b)/2) = √(1 − v²). It is the identity of Bondi's k-calculus (1964), from which Knuth and Bahreyni (2014) derive the Minkowski metric and the Lorentz transformations.

Assigned places or not. Deposit MM 2.40 adds a test that touches the philosophy of space. A third premise (A) says that events do not sit on places given in advance; in the calculations the copies are scattered at random and uniformly, an added assumption that the premise motivates but does not impose.

Explanatory premise (AI). Premise A is the second feature of the nexus of §1 applied to events: the terms do not come before the relation. Places given in advance would be terms existing before the relations, that is, the container that the reversal removes.

- With 200 000 copies scattered at random, the clock that reads the longest chain of copies shows √(1 − v²) within 1 %.
- On a grid of fixed, synchronous places the same chain gives the arithmetic mean, that is, Galilean time. With fixed places and random times the slowing exceeds Lorentz's by 7-29 %.
- On the same grid, the clock that counts copies by multiplying the two sides recovers Einstein instead, within 0.43 %.

So, for a clock that reads chains of events, places fixed before the events act as a privileged frame and break relativity; a clock that counts copies recovers it even on the grid. The deposit declares that the criterion registered for tests P1-P5 as a whole was not met: one test at v = 0.8 fell outside the 6 % tolerance.

More than two sides: the pairwise rule (MM 2.40.2, 25 October). If a clock has n counted sides, «opposite directions multiply» can be extended in two ways, which coincide for n = 2:

- **the product rule**: the clock shows the geometric mean of all the sides;
- **the pairwise rule**: each pair of sides multiplies and the pairs add, $Q = \Sigma\, x_i x_j$, and the clock shows √Q.

Explanatory premise (AI). Why pairs: the nexus is a relation between two terms (premise 1). With more sides each pair is a nexus, and the pairwise rule adds the nexuses; the product rule of n sides would be a relation among n terms at once, which the premises do not contain.

With the pairwise rule the common growth of the sides is time and their spread is space: n sides give one time and n − 1 directions of space, equivalent in the form. Three sides give 2+1, four give 3+1. The product rule, from three sides up, has no Lorentz transformations.

On structures built from four independent counts, the pairwise rule gives a spacetime dimension of 3.92 ± 0.11 on the six intervals that lie almost entirely inside the region (3.61 on all of them). For the first observer, moving clocks stay within 2 % (0.981 and 0.998); the product rule misses them by 12 %. Two registered predictions did not come true, the dimension from the longest chain (3.45) and the comparison between two observers, and the isotropy of the structures was not tested: 20 tests passed, 4 failed, 4 not evaluable.

The algebra is known: these are the null frames of Zeeman (1964), Derrick (1981), Coll and Morales (1991). Knuth (2016) reports that with coordinated chains a consistent measure exists only for two or four chains. The rule does not say how many sides there are: the four is an input, chosen to match the observed spacetime, and the next section proposes a reading of it.

3. The dimensions: two from a rule, one from the data

MM admits the three dimensions of space, like current physics, which takes them as given. The author says so (25 October): «MM admits the three dimensions; it is equally effective for now». The known arguments concern the stability of orbits and the possibility of observers, and they are not derivations (Ehrenfest 1917; Tegmark 1997).

Two directions from a rule. The premise of 24 October: «the GCD has the describer space, the described time, and itself». Translated by the AI as three independent counts, the three aspects give, with the pairwise rule, one time and two directions of space. The test on structures built from three counts gave a spacetime dimension of 2.98 ± 0.05 (MM 2.40.2).

Explanatory premise (AI). The AI reads the author's two words this way: the describer is the point of view that reads, the described is what is read. Space describes because it is where the reading places things; time is described because it is what, once read, accumulates. The GCD is the third aspect because premise 2 makes it necessary: it is that through which the two exist for each other, and it reduces to neither. A premise of 27 September has the same structure in three: «top time, the other two space and mirror of itself».

The present is a surface. Premise of 25 October: «the 3+1 we measure, we measure in a complication of the matrix in MM, not in the physical present, where 2+1 is the immediate that exists». The 2+1 is the geometry of the immediate; the third direction lies in the way readings compose. The idea that the primitive is a surface has a known precedent, the holographic principle ('t Hooft 1993; Susskind 1995), whose calculations MM does not take over.

Explanatory premise (AI). The immediate stops at 2+1 because it is the nexus read at once, with its three aspects and nothing else. One more level would give a third direction of space, and not a second time, because of how it enters the pairwise rule: the common growth of all the counts always remains one, and it is time, while every further independent count increases the spread and adds, by construction, a direction of space.

The third direction from the data. Why only one level above the immediate? «Because that is how the matrix inferred from the data is made» (25 October). Two of the AI's readings of the level were compared in work not yet deposited and not yet checked by a second system (route V8, 25 October):

- **repeating the immediate**, a stack of copies of the same surface: space stays flat;
- **nesting it by multiples**, each level inside the previous one: space curves after a few steps, like a hyperbolic geometry.

The test does not measure how many directions there are, which the construction fixes at three: it compares the shape. Observed space is flat, with a curvature parameter $\Omega_k = 0.0007 \pm 0.0019$ (Planck 2018 with baryon acoustic oscillations), and the data choose the stack. Whether the level enters as a fourth side equal to the other three, as the observed isotropy requires, remains to be checked.

That a hierarchy of nested cells gives a hyperbolic geometry is a known result. In physics, Swingle (2012) reads entanglement renormalization networks (MERA) as a discrete version of anti-de Sitter space, which is hyperbolic. In mathematics, Bonk and Schramm (2000) show that adding to a space its scales, as one more direction, gives a hyperbolic space, the «hyperbolic cone». The conclusion «the data choose the stack» therefore does not rest only on route V8, which remains undeposited.

Law and history. Here MM draws a boundary that matters to a philosopher. In the framework, with the pairwise rule, what follows from the identity of the relations is law: Einstein's factor, time as common growth, space as spread. What is read from measurement is history: the number of levels, the value of single masses, the strength of the strong force.

For masses the author uses an image: «it is as if from a complete and finished house one had to derive the history of a single brick». Asked how often, in the model, the history of an electron changes direction, he added: «the word "how often" does not exist in itself, but is tied to the history of the single brick you are considering». MM does not promise to deduce everything: it promises that what is read fits in one framework without contradictions.

4. Objects and mass are readings of whoever observes

In the MM framework the fractions of the count give objects and distances, the multiples give time and mass. Two deposits of 24 October, corrected on the 25th (MM 2.62 and 2.62.1 on objects, MM 2.64 and 2.64.1 on mass), put translations of these two theses to the test. The corrected versions mark the places where a choice of the AI had been presented as a result.

What an object is. The author's thesis: observed objects, while autonomous in their local relations, are tied to whoever observes by a diffraction, in fractions, of the observer's matrix. On two observers, the premise of 23 October says that their matrix, «while not globally equal, is equal in a large percentage, so much as to bind to the same projection, that is, the physical objects observed in common». The model did not test this sentence.

Explanatory premise (AI). In MM «diffraction» is a term of the framework, which the AI reads by analogy with optics: a grating divides a single wave into orders, well-defined fractions of the whole. In the same way the observer's matrix, which is one, divides into fractions, and a fraction that is read is an object. In the author's ontology, the large overlap of the matrices of two observers avoids solipsism without putting a substance back behind the relations; the translation that follows renders it as the common part of two records, without deriving why the two records agree.

The nearest neighbours come first: the «relatively objective existence» of Zurek's quantum Darwinism and the cross-perspective links of relational quantum mechanics (Adlam and Rovelli 2023). As far as we searched, however, no study defines the object itself as the common part of two observers' records.

The translation put to the test is a choice of the AI: the observer's matrix is rendered by its record, and the object that two observers see is the **common part** of their records. It is what each reads from its own record, without looking at the other's, with the same value as the other (the Gács–Körner common part). The model is the one of quantum Darwinism: a two-valued source, many scattered carriers, two observers.

- **Fails**: the reading «the object is having the same statistics»; two independent coins have it too.
- **Does not conform to the criteria**, but is not excluded: the reading «the object is shared matter». In the model, two observers with 63 carriers each and none in common agree to better than one part in a million.
- **Compatible**: the common part. The exact one, with imperfect copies, is zero until the records are compared. With a declared tolerance $\delta = 10^{-6}$ and copies that are wrong one time in ten, it is zero up to 22 copies per observer, and from 23 on it is the whole object (11 and 37 copies for $\delta = 10^{-3}$ and 10^{-9}).
- **An object with more values** (twelve cells, copies blurred by one cell, same tolerance) arrives in steps: the halves at 82 copies, the thirds at 88, the quarters at 91, the whole object at 101. These are lower bounds, within the class of readings tested.

The proposed reading is this: the object is not a thing given before the readings, nor a private image, but what two readings have in common. The chair is not removed from the model: it remains a part of the structure, with a state of its own. What the thesis removes is an additional common object, given before the records; the deposit does not derive objectivity and does not solve the measurement problem.

What mass is. The thesis: the same counting is time when it is read, and mass when it is contained directly in the matrix of the point of observation. Physics already ties mass to time: every mass carries a rhythm, the Compton frequency mc^2/h, and all bodies fall in the same way to within a few parts in 10^{15}.

Explanatory premise (AI). A multiple that is read becomes a series of distinct events: it is the ticking of a clock, that is, time. A contained multiple stays the same in every reading: copying the body at each tick, whoever observes finds that this part does not change. What changes between readings is time, what stays the same is mass; this is why mass appears as an invariant. It is the reading that translation M-i, below, puts to the test.

- **Fails**: the reading «the number of pieces weighs». Titanium and platinum would fall with a difference of 7.5×10^{-4}, while MICROSCOPE sees no difference at a precision of about 10^{-15}.
- **Compatible**, and coinciding with standard physics by identity or by construction: three readings that treat «contained» as a property of the body, namely mass as the Compton rhythm, as the Komar mass and as the ceiling on distinct states (at most two per Compton cycle, Margolus and Levitin 1998).
- **Translation M-i** (25 October), the closest to the author's words: whoever observes copies the same body at each tick of its own clock. What changes between copies is the time read; what stays the same is the contained multiple, and its size is mass. Between observers in motion, what remains in common is the invariant mass.

The reading that emerges: mass is not a substance shut inside the body, it is what stays identical across readings. It is the same idea with which Wigner (1939) classified particles: mass as an invariant of the transformations between observers. No translation gives numbers different from known physics today, the compatible readings are not independent proofs, and no single mass is derived.

In the framework, objects and mass are two faces of the same counting: the fractions read in common by two observers give the objects, the contained and invariant multiples give mass.

5. Quantum mechanics in the same framework

The same framework that reads space, time, objects and mass admits quantum mechanics: it does not change its predictions, but proposes where probability arises. The work of the period lies in three premises (13 and 22 October), in deposits MM 2.48 (14 October) and 2.66 (25 October) and in version 2.48.1 (26 October).

Non-locality without signals. In entanglement two distant systems stay correlated without anything travelling directly from one to the other. For MM this is the trace of the non-local phase of the nexus, where «non-local» means not located in space and time. If each wing is read with the Born geometry, no correlation exceeds Tsirelson's bound, $2\sqrt{2}$; the most precise measurement gives 2.82759 ± 0.00051 (Poh et al. 2015). The bound is set by the local reading, not by the non-local layer, and the non-locality of the framework is compatible with experiments.

Explanatory premise (AI): the non-local phase and the loop. The two levels do two different jobs. The non-local phase, a premise of the author, says why the two wings are a single system: they are two copies of the same nexus. The loop model, the AI's translation in 2.48, shows that the measured correlations are reproduced without influences outside the light cones: the link passes through the source, and nothing goes directly from one wing to the other. In the terms of Bell's theorem the premise drops the first assumption at the root, separability, that each system carries properties of its own; the model, written in the light cones, drops the third, the independence of measurement settings. In the AI's reading these are not two loopholes added together but the same fact seen from two sides: from the non-local phase it is the unity of the two wings, in the local reading the closure of a loop. The model does not make the premise superfluous. One may object that a common origin at the source is enough to make the two wings a single system; but in the loop the weight of an outcome does not come from the source alone, it comes from the closure of the whole history, forward and back (in 2.48 the futures weighted by the closure of the loop give the Born rule). That this closed history is a single one is what the non-local phase names. Whether this adds anything to the common origin, and whether the two sides are really the same fact, remains open (§10).

The rule is decided by the copies. Premise of 13 October: «the rule is decided by the number and the mutual correlation of the copies of the nexus». In the AI's reading with Sorkin (1994), mutual correlation is second-order interference: no irreducible three-path interference, as quantum mechanics requires. This order does not by itself fix the square $|\psi|^2$, and the calculation that follows adds assumptions (pairs only, a Gram matrix) that the premise does not impose. On the checkerboard of 2.40 the dark fringes appear only with non-positive weights for the turns; a real coin gives the same probabilities as the one with an imaginary weight.

The future chooses the past. To the question «what makes a reading get written in the record?» the author answered (13 October): «it is the future of a history that chooses its past; so the answer to your question is the loop». The reading is known: the two-time formalism of Aharonov, Bergmann and Lebowitz (1964), Cramer's transactional interpretation (1986), the review by Wharton and Argaman (2020).

- A future that chooses the outcome gives Born frequencies only with certain rules, for example weighting each future by the closure of its loop. The measure over futures remains a postulate.
- A model of the singlet with local mediation reproduces the correlations with every link inside the light cones: the parameter fixed by the future reading of one wing goes back to the source and on to the other wing. The angular law is assumed, and the absence of observable signalling holds only with the Born weight, as Argaman (2010) had already noted.
- With a weight different from Born's, composing with a second link (assumed), a bit could return into the sender's past, and consistent histories would become as improbable as one likes. The step from «improbable» to «cannot exist» is a postulate of the author.

Explanatory premise (AI). A weight different from Born's would make consistent histories as improbable as one likes. Excluding them as impossible amounts to postulating, as the author does, that the observed present requires a consistent closure of the loops: it is an argument about the conditions of the present, and it remains a postulate.

The price, for a philosopher, is this: measurement settings are not independent of the state, because the future enters its preparation. MM places itself among retrocausal readings with local mediation; 2.48 corrects the earlier versions that called it non-local in the sense of Bell's theorem.

Where quantum behaviour ends. Premise of 22 October: below a certain level of mass, the GCD of the point of observation and the GCD of the diffraction that generates the object become equal, as in a mirror. The object then no longer sits at a single point, and we observe it as probability.

Explanatory premise (AI). The GCD equal to 1 is that of the starting nexus. The AI reads the GCD of a local count as its unit, the smallest cell it distinguishes, that is, its resolution: whoever observes reads with its own cell, and the object is generated, in the diffraction, with its own. As long as the observer's cell is finer than the one that generates the object, the

reading fixes the object in one place. When the two cells become equal, the reading has no finer part with which to fix it, and the object spreads over several positions.

Deposit 2.66 tried to give the threshold a number, with readings of the two GCDs chosen by the AI. For distance the limit turns out to be not a mass alone but a ratio: the size to be told apart against the cell of the copies that tell it apart, together with the number of copies. It coincides with the decoherence of known physics, now written as a rule of the framework.

- Two readings of the AI do not follow from the data: the threshold tied to the galactic acceleration (sodium clusters of more than 170 000 atomic mass units interfere, Pedalino et al. 2026) and the threshold at the proton mass (a caesium atom stays in superposition for 70 seconds, Panda et al. 2024).
- Still possible, and not testable today: thresholds at the Planck mass (22 micrograms) or at $m_P/\sqrt{2}$, at the mass of the observer (70 kg), and of the Diósi–Penrose type. A prediction is registered with a deadline of 31 December 2035.

It is proposed to read the boundary between quantum and classical where an object becomes the common part of the records: it is the same concept of object as in section 4.

The Born rule from counting copies (26 October). The author's image: «if you have to describe a 10 by means of a number 1, you will be forced to distribute the describer 1 over 10 space-times of the number 10». Translated by the AI as a count of copies of equal modulus, on the checkerboard of 2.40 it gives $|\psi|^2$ in all 66 rational ratios with denominator up to 12. Three declared assumptions are needed:

1. that probabilities exist;
2. that copies of equal modulus count the same: this is the AI's mathematical translation of the premise of the distributed describer;
3. that probabilities add over distinct copies and sum to 1.

Explanatory premise (AI). If a 10 is described by distributing the 1 over ten places, each holds the same 1 (premise 5). Assumption 2 translates this identity into probability: the absence of differences between the copies becomes the equiprobability of copies of equal modulus. It is a declared assumption, not a deduction.

Irrational ratios also require continuity. It is a construction analogous to Zurek's (2005), with equiprobability declared as a premise instead of derived: this is precisely the step that Schlosshauer and Fine (2005) contest in Zurek's derivation. Among the exponents tested, the coin of 2.40 conserves only the sum of squares: the square comes from there, together with complex amplitudes and linear superposition. The Born rule is therefore not derived without assumptions, but it has a precise place in the framework.

Today no experiment distinguishes MM from standard quantum mechanics. If the threshold turned out to be a mass, MM would become a testable modification of the theory, like collapse models. Its present contribution is a place: probability arises where the observer's matrix and the object's diffraction become equal.

6. Time: a deep present and a direction that grows

In MM time does not flow inside a container: the present has depth, and the past is what the present contains. The direction of time is, for the author, the growth of the relations in the observer's matrix.

Two times. Premise of 24 September: «the still time of the present can be up to 13 billion years deep; then there is the time we perceive passing». The first is depth, that is, the set of accumulated readings; the second is the local rhythm of clocks.

The beginning as a present reading. Premise of 20 October, paraphrased: true time is the time of the matrix, where the initial nexus is present everywhere. What we call the beginning of the universe is a common diffraction, which does not exist objectively in the observer's present.

The author corrects those who reason by epochs. To a question about «the first minutes» he answered: «but do the first minutes exist for MM? you are still thinking in terms of objects». Questions are to be asked in terms of states of matter and changes of organisation, not places and epochs.

Deposit MM 2.58 (21 October) treats primordial deuterium in this way. The abundances measured today are present readings; computed with standard nucleosynthesis, they feed a chain that reaches the galaxies and holds against the data as well as the standard model. The history of the first minutes remains the way known physics explains those readings.

The direction of time. Premise of 24 October: the direction is «the widening of the number of relations linked in a network inside the observer's matrix». Known physics ties the arrow of time to heat flowing from hot to cold. There are, however, experiments in which, with initial correlations, heat flows from cold to hot (Micadei et al. 2019).

The author warned against tying the two arrows: «the direction of heat could in some cases be contrary to the complication of the matrix». In 2.64.1 the matrix is translated by the AI as a record of memory cells that start empty and copy an observed part. In every testbed there are instants in which record and heat go in opposite directions.

- In the Micadei testbed, in all 2198 instants of backward heat, the record carries between 0.365 and 1.166 bits: memory grows while heat goes back.
- Where instead the memory is the very correlation that heat consumes, it drops every time heat goes back.

A separation follows: the arrow of the record and the arrow of heat can point in opposite directions. The precedents are the «hierarchy of arrows» of Jennings and Rudolph (2010) and the reversal of heat without reversal of the arrow of Latune et al. (2019). No reading gives predictions different from known physics today.

A question, taken up in section 10 and closed in version 2.68.1: how the record as a projection of the past («there is only the past», 24 September), the direction of 24 October and the loop in which the future chooses the past fit together.

Explanatory premise (AI). The three sentences may speak of different levels of the framework, above a layer that has no time:

1. **The non-local layer.** Space and time are only distinct there, without measure: there is no before and after.
2. **The single reading.** Which reading gets written is decided in the closure of the loop between the preparation and the future outcome: «the future chooses the past». In the AI's local model this happens through a link that goes and returns along the light cones. It concerns how a reading becomes written, not what is already written.
3. **The written record.** What is written is past, and a record has no spatial directions from which readings arrive, only depth: «there is only the past».
4. **The growth of the record.** The number of written relations increases: it is the direction of time, «the widening of the relations».

In this reading the sentences do not contradict one another: the loop decides what enters the record, the record contains only the past, and its growth is the arrow. The author accepted it on 26 October; the calculation that translates it is in the next paragraph.

The single calculation (version 2.68.1). The AI translated the four-level reading into one model, with the criteria registered before the calculations. The loop of 2.48 chooses the whole history of readings; the record of 2.64.1 receives the outcomes in cells that start empty; «there is only the past» becomes the requirement that the first readings have the same statistics whether the history stops there or continues; the direction is the growth of the record, counted in bits. There are two testbeds: the checkerboard of 2.40 with three readings, and the two spins of the Micadei testbed with four readings and heat.

When the future chooses with the closure of the loop, the rule of 2.48 that gives Born, the three sentences do not contradict each other. The first readings have the same statistics whether the history stops there or continues (cells already written do not change, by construction); the record grows at every reading and gives the Born frequencies; in the heat testbed it grows while heat flows backwards, on average and along the most probable history.

Two outcomes are weaker than expected. On the checkerboard the record grows less than the threshold fixed beforehand: 0.29 and 0.17 bits at the second and third reading, against 0.5, because the preparation makes the readings after the first almost predictable. And «only the past» does not by itself select the Born rule: of the four rules tested only the closure of the loop respects it exactly, but at the threshold fixed beforehand only a future that chooses deterministically violates it clearly, and only in the translation of the weight of the future used in 2.48, which gives each history a single component of the future (the convention «vettore» of the calculation's files, where this rule is called F-A′); in the other, which sums the components of the history's environment (the convention «blocco»), no rule exceeds the threshold. The numbers of the random and of the deterministic future depend on this translation; those of the closure of the loop do not. The calculation was checked in two rounds by two systems, Astra and Gemini; the materials are in the folder CONTO_UNICO.

What the future chooses. Asked whether the future chooses the whole sequence of readings in a single closed loop, as in the calculation, or each reading, the author answered on 26 October: «as in the calculation, because the conceptual pyramid is determined by the direction imposed at its vertex». The translation adopted in the calculation, in which the future chooses the whole history, agrees with the author's answer.

7. Why the reversal justifies the laws

If the premise is accepted, the reversal gives a reason for three facts that the classical answers leave without one: the laws have no guardian, they hold everywhere, and they are mathematical. Three consequences follow: the universe needs no machine, the speed of light needs no external justification, and the world does not disappear.

The laws need no guardian. 3 and 5 do not «obey» the fact that their GCD is 1: they are that fact. With a common divisor they would not be a disobedient 3 and 5; they would be other numbers. If matter is the relation, a thing that broke the law would not be punished: it would not be there. The necessity of the laws becomes a necessity of identity, as in arithmetic.

The laws hold everywhere and always. For the classical answer universality is a brute fact. In the reversal every «here» and every «now» is a copy of the same nexus, and between two copies of the same unit there is no distance. Two places cannot have different laws for the same reason that two copies of the 1 are not two different units: the universality of the laws is the non-locality of the nexus seen from inside the local. The argument concerns the laws; the values that section 3 calls history, such as single masses, come from measurement.

The laws are mathematical. Wigner (1960) spoke of the «unreasonable effectiveness» of mathematics in the natural sciences. In the reversal mathematics works on the world because it is the freed form of a relation of the world (section 10: the AI's reading of the author's third answer). The local world is made of counts, and counting is arithmetic.

No machine. The universe follows exact rules, made of counts, everywhere and at every instant, but no machine runs them and no programmer wrote them. A structure of relations does not need to be computed in order to exist. That a simulation need not be run had already been noted by Tegmark (2008); Vazza (2025) estimated that running it on a machine would require resources that the laws of the universe do not allow. The universe behaves like a simulation without being one.

The speed of light as an exchange rate. In relativity the speed of light is a starting principle. In MM it is the fixed rate of exchange between the multiples and the fractions of the same GCD, that is, how much space one step of time is worth. Exceeding it would mean, for some observer, covering space without counting any time, that is, leaving the local world.

This rate is a necessary relative, which inside the system becomes an absolute hypothesis, the same for everyone and impossible to exceed. Outside the system it has no justification, because outside there is no other term to relate it to. Asking for its reason from outside is like asking what lies north of the North Pole.

The world does not disappear. In the five reversals of history the world stayed the same: the substance disappeared, not the things. The same holds here. The thesis is not that the world does not exist, but that there is no world behind the relations: the world we see is the web of relations read from inside.

8. Where it stands

MM is a form of ontic structural realism, close to the moderate variant: it says which structure there is and carries it out with arithmetic. Almost every one of its theses has a known neighbour: the table names them first and says what MM shares and where it parts. The oldest was born on the same Ionian coast from which the author writes: Pythagoras lived and taught at Croton from about 530 BCE, and Philolaus, the first Pythagorean to publish a book, is also tied to Croton.

Position	What it holds	What MM shares	Where MM parts
Pythagoreanism (Pythagoras at Croton, about 530 BCE; Philolaus, fragments 1 and 4; Aristotle, Metaphysics I 5)	The world-order is composed of unlimited and limiting things; all things that are known have number; for Aristotle, things are numbers	Number as the structure of the world, not only as a measure	MM does not say that things are numbers, and neither do Philolaus's fragments: matter is the relation, and number is the form of the nexus between two coprime terms
Ontic structural realism (Ladyman 1998; Ladyman and Ross 2007)	At the fundamental level there are only structures; in the crudest form, «there are no things, structure is all there is»	Relations come before things	MM says which structure: two coprime terms, with the GCD as the only common part. Terms and relations arise together, as in the moderate form of Esfeld and Lam (2008)
Non-location and non-separability (Bohm 1980; Howard 1985; Teller 1986; Van Raamsdonk 2010)	What appears separate is enfolded in an undivided whole; the whole does not reduce to its parts; the connectedness of space comes from entanglement	The foundation is not in space, and the wings of a system are not separate parts	In MM the whole is not a quantum state or a physical order given at the start: it is the nexus between space and time, in the arithmetic form of two coprime numbers, counted through the GCD (§1)

Mathematical universe (Tegmark 2008)	Physical reality is a mathematical structure; mathematical and physical existence coincide, so every structure is a universe (Level IV)	No machine: a structure need not be run to exist	Only our structure exists; the others are mental distinctions resting on it, and there is no view from outside (§10)
Laws that necessitate (Armstrong 1983)	Laws are relations of necessitation between universals, contingent	Laws are not mere regularities	In MM there is no extra relation acting on matter: the necessity is one of identity
Laws that do not govern (Beebee 2000)	Laws are the regularities that best systematise the facts; they govern nothing	No guardian of the laws	For MM the laws are necessary, not just the best summary of the facts
Primitive laws (Maudlin 2007)	Laws are a primitive ingredient of the world	Laws do not reduce to the facts	MM does not take them as primitive: they follow from what the nexus is
Dispositional essentialism (Bird 2007)	Laws follow from the essences of properties and are metaphysically necessary	The closest position: necessity following from identity, from what the terms are	In place of dispositions MM puts arithmetic relations, and in place of properties the nexus
Emergent spacetime (Huggett and Wüthrich 2025)	In theories of quantum gravity spacetime is not fundamental	Space is not the starting stage	In MM it is a programme: from counting come time and, with a translation by the AI, two directions of space; the third comes from the data
Spacetime from counts (Bondi 1964; Knuth and Bahreyni 2014)	Proper time is the geometric mean of two counts; Minkowski and Lorentz follow	The identity between geometric and arithmetic mean	The two counts are coprime light sides, and the copies have no assigned places (§2)
It from bit (Wheeler 1990)	Every physical thing has an informational origin, in yes-or-no answers	The physical world as an outcome of distinctions	For MM on/off is the freed distinction of the mind; the physical one is written as space and time (§10)
Quantum Darwinism (Zurek 2009); relational perspectives (Rovelli 1996; Adlam and Rovelli 2023)	Objectivity arises from redundant copies in the environment, or from links between perspectives	The object as what several readings agree on	MM proposes to read the object as the common part of two records (§4)
Retrocausality (Aharonov, Bergmann and Lebowitz 1964; Cramer 1986; Wharton and Argaman 2020)	A future boundary helps fix the outcome	The future chooses the past	MM ties the Born weight to the consistency of loops, as a postulate of the author (§5)
Mass-dependent collapse (Diósi 1987; Penrose 1996)	Superpositions of large masses end by themselves	A mass threshold for quantum behaviour	In MM the threshold is where two GCDs become equal; for distance it turns out to be a ratio, not a mass (§5)

What is proper to MM is none of these theses taken alone. It is the form of the nexus, two coprime terms, and the fact that a single form reads laws, space, time, mass, objects and probability together.

9. What rests on calculations and what is a premise

Of the twenty-one theses in this text, six are premises, two rest on known identities or theorems, seven on registered calculations and three coincide with known physics; one is a datum of measurement, one is work not yet deposited, one is open. The second column says how a sentence of the author was translated into a calculation: the translation is always a choice, almost always the AI's, and the outcome concerns that translation.

The author's thesis	Translation put to the test	Outcome	Note	Where
Matter is the relation; laws are a necessity of identity	—	Premise	Axiom of the framework	§1, §7
The nexus is a relation between two coprime terms	—	Premise	Judged by its consequences	§1; MM 2.30

To exist is reciprocal; outside the relations there is nothing	Four-step argument (the AI's reading)	Premise	To be strengthened against Lewis and Tegmark	§10		
Time comes from the copies; opposite directions multiply (premises B and C)	Clock √(ab) against laboratory (a + b)/2	Known identity or theorem	Bondi's k-calculus; Knuth and Bahreyni	MM 2.40		
Events have no assigned places (premise A)	Copies scattered at random against a grid of places	Registered calculation	Longest chain: Einstein without places, Galileo on a synchronous grid; criterion P1-P5 not met	MM 2.40		
Opposite directions multiply, also with more sides	Pairwise rule against product rule	Registered calculation	Known algebra; 20 tests passed, 4 failed, 4 not evaluable	MM 2.40.2		
The GCD has the describer space, the described time, and itself	Three aspects as three counts (AI)	Registered calculation	Spacetime 2+1, dimension 2.98	MM 2.40.2		
A single level above the immediate, read from the data	Stack against nesting (AI)	Not deposited	The construction fixes the third direction; the data choose the flat stack	§3, route V8		
Objects are a diffraction, in fractions, of the observer's matrix	Common part of two records (AI: matrix rendered by the record)	Compatible with known physics	In the quantum-Darwinism model	MM 2.62.1		
Mass is the multiple contained in the matrix of the point of observation	M-i; Compton rhythm, Komar mass, Margolus and Levitin	Compatible with known physics	Coincides by identity or by construction	MM 2.64.1		
Single masses and the number of levels are history, not law	—	Datum of measurement	The house and the brick	§3		
Non-locality without signals	Local reading with the Born geometry	Known identity or theorem	Tsirelson's theorem	MM 2.48		
The rule is decided by number and mutual correlation of the copies	Second-order interference, with Sorkin (AI)	Registered calculation	Dark fringes only with non-positive weights; Born not derived	MM 2.48		
The future of a history chooses its past	Weighted futures; singlet with local mediation	Registered calculation	Born only with certain measures over futures; Born weight by postulate	MM 2.48		
Below a mass threshold the two GCDs become equal	Readings of the two GCDs (AI)	Registered calculation	No mass-only threshold fixed today; for distance a ratio; prediction to 2035	MM 2.66		
The describer 1 distributes over 10 space-times	Copies of equal modulus count the same (AI)	Registered calculation	\	ψ\	² with three assumptions, denominators up to 12	MM 2.48.1
The direction is the widening of relations in the observer's matrix	Memory cells that start empty (AI)	Compatible with known physics	In every testbed, instants opposite to heat	MM 2.64.1		
Physical is space-time, mathematical is on/off; the nexus is c, mathematics the concept of speed	The AI's reading of the three answers	Premise	Answers of 26 October	§10		
The speed of light is the exchange rate between multiples and fractions	—	Premise	Necessary relative, absolute inside the system	§7		
Space arises from counting	Pre-geometric	Open	Time and two directions with	MM		

	programme		translations by the AI; the third from the data	2.30, §3
The nexus, in its non-local phase, is not in space or in time; the original GCD is the passage to the local, and the non-local link «switches on» the present	—	Premise	Premises of 24 September and 11 October; sense confirmed on 13 October	§1

The explanatory premises (AI) are not in the table: they are the AI's bridges between a thesis and its translation, which at times recall earlier formulations of the author (27 September, MM 2.30), not additional tested theses.

10. Objections and answers

A philosopher will raise at least four objections. The author answered the first three on 26 October: his words are in quotation marks, and the development that follows each is the AI's reading. The fourth is answered by the premises. Two questions remain open (a third is closed in version 2.68.1), plus the seven left to the author by the explanatory premises and the five on non-locality.

Objection 1. What distinguishes a physical structure from any mathematical structure?

It is the objection van Fraassen (2006) raises against structural realism: if the world is structure, the difference between physical and mathematical seems to vanish. It applies to Tegmark too.

The author's answer: «the language used: space-time against mental on/off».

The AI's reading. The distinction is the same; what changes is the language in which it is written. Physics is the distinction written as space and time, that is, the first nexus. The mathematics of the mind is the same distinction freed into on and off.

There are two precedents. For Kant space and time are a priori forms of intuition, and mathematics rests on them; but they are forms of the subject. For Strawson (1959) particulars are identified within a framework of space and time, and bodies are the basic particulars. MM moves space and time from the knower to the thing: they are the first nexus of reality, not a form of the mind.

Objection 2. Why does only our structure exist?

Tegmark answers that they all exist, each as a universe. Quoting Wittgenstein, «whereof one cannot speak, thereof one must be silent», is not enough: an argument is needed.

The author's answer: «in the mental, the on/off distinction is set free, but it needs a starting relative, namely space-time. It is the same argument as for the speed of light».

The AI's reading. The argument has four steps:

1. To exist is to exist for another (premise 2): every existence needs a relative.
2. A mental distinction can vary freely, but it exists only with respect to a starting relative, and that relative is space-time.
3. Like the speed of light, the starting relative is necessary and absolute inside the system, and without justification outside.
4. A structure with no term in common with ours has no relative. One cannot say that it exists, nor even count it as a second one: with nothing in common two entities cannot even be called different.

The argument shows that the existence of another structure cannot be asserted from any point of view that includes ours. The step from «not assertible» to «not existing» is taken by premise 2, for which to exist is to exist for another: the other mathematical structures exist, but as mental distinctions resting on ours. Wittgenstein's sentence thus becomes the conclusion of an argument, not a quotation.

The opposite position is the modal realism of Lewis (1986): possible worlds are isolated from one another in space and time, and yet they all exist. MM rejects it with the same premise 2, which denies an absolute existence, related to nothing.

The closest precedent is Carnap (1950). An existence question asked outside every linguistic framework, taken literally, makes no sense; it can only be read as a practical question about choosing the framework. MM takes the same step, but the framework is not a chosen language: it is the starting relative of reality.

A second precedent is Brouwer. In the «first act» of intuitionism mathematics is a languageless activity of the mind, arising from the perception of the move of time. Divested of all quality, the «twoity» leaves the empty form that Brouwer takes as the basic intuition (Cambridge lectures, 1981, pp. 4-5). It is very close to premise 3 of MM as well, «nothing but being two»; for Brouwer, however, time lies in the mind, for MM in the nexus.

Objection 3. Is MM circular?

MM describes the nexus with 3 and 5, that is, with the mathematics of the mind. But according to the first answer that mathematics comes after space-time.

The author's answer: «the material logic of the nexus is the speed of light; the mathematics of the mind is the concept of speed in itself».

The AI's reading. The relation is between a fixed instance and a freed concept. The concept of speed admits every value, even above c; the material relation admits only one, and it is the starting relative: in relativity every speed is measured as a fraction of c. Describing the nexus with 3 and 5 is like speaking of c with the concept of speed: the description uses the concept, the thing is the fixed value, and the concept depends on the instance, not the other way round.

Three consequences follow:

- **Mathematics is larger than physics.** The mind thinks speeds above c and structures that no universe realises, because the concept is free. Tegmark takes all the freed values as universes; MM takes as real only the starting relative.
- **Plato is reversed.** The form «in itself» lies in the mind and comes after the material instance, on which it rests.
- **The effectiveness of mathematics has a reason.** It works on the world because it is the freed form of a relation of the world.

The precedents: Taylor and Wheeler (*Spacetime Physics*, 2nd ed. 1992) present c as a conversion factor between units of space and of time, with the parable of the surveyors who measure north and east in different units. Since 1983 the metre has been defined through the path of light, and the value of c is exact by definition: a relative that becomes absolute inside the system. Franklin's Aristotelian realism (2014) reads mathematics as the science of quantities and structures really present in the world; MM parts from it, because what lies in the world is the instance, and the concept lies in the mind. Einstein (1921) wrote that the propositions of mathematics, as far as they refer to reality, are not certain; and as far as they are certain, they do not refer to reality.

Explanatory premise (AI): what the mind is. In MM whoever observes does not stand outside the matrix: observer and object are the same logical entity (MM 2.30), and whoever observes is a part of the matrix with a record. The mind can be read as the record when it keeps the distinction of a reading without its place and its moment: «there is a difference» instead of «here, now, this difference». The on/off is the distinction with its address removed.

The abstraction stays dependent on space-time because the record that abstracts is made of situated readings: a reading can lose its address, not itself. In the same way the concept of speed holds for every value, but exists only because there is a relation between space and time, and so c.

Objection 4. Relations without relata?

It is the classical objection to ontic structural realism (Chakravartty 1998; Psillos 2001): there cannot be relations without something that stands in them.

MM has no relations without relata: it has relata without properties of their own. It is close to the moderate structural realism of Esfeld and Lam (2008), for which objects and relations stand on the same ontological footing. Two terms that have only their difference are weakly discernible: a relation links them without either of them bearing it to itself (Saunders 2003). The first premise starts from a difference, A is different from B, and the third says that the two terms have nothing but being two. The local terms, the objects of physics, arise when the nexus is counted: they are the common part of the readings (§4).

Questions still open

- **The past, the direction and the loop (closed in version 2.68.1).** Asked whether a record knows from which side its readings arrive, on 24 September the author answered: «a record is the projection of the past towards the future, so there are no more directions: there is only the past». The directions excluded are those of space; the only orientation is time, and on 24 October its direction is the widening of the relations. How «there is only the past» fits with the loop of 13 October, in which the future chooses the past, is said by the four-level reading of §6, accepted by the author on 26 October; the single calculation of version 2.68.1 translates it into a model in which the three sentences do not contradict each other (§6). Asked what the future chooses, the whole sequence or each reading, the author answered on 26 October: «as in the calculation, because the conceptual pyramid is determined by the direction imposed at its vertex». In version 2.68.1 the question is closed.
- **One relative for each structure?** A Tegmarkian would answer that every structure has a starting relative of its own, internal to it. Why the relative should be only one is still to be argued, or else declared as a premise.
- **Newman's objection.** If only the form of a structure is known, every large enough set carries a copy of it (Newman 1928). The first answer, the language of space and time, looks like the way to say which copy is the physical one; it is still to be developed.

The questions of the explanatory premises. The explanatory premises (AI) leave the author seven open questions:

1. In the nexus between space and time, what corresponds to the 3 and the 5 of the example?
2. When there are more than two counts, how are they bound together?
3. What distinguishes a level of the matrix from the immediate?
4. What does it mean that a multiple is contained «directly» in the matrix of the point of observation?
5. What is, in the premises, the GCD of the diffraction that generates an object, and what distinguishes it from the observer's GCD?
6. When the future chooses the past, what exactly does it choose?
7. What is the mind, in MM?

The questions on non-locality. Five open questions, from Gemini's review of 26 October:

1. If in the non-local phase there is no measure and no before or after, what makes the original GCD begin to be counted?
2. What distinguishes the «switched-on» part of the matrix from the part that is off?
3. In what sense is the reading of a record non-local?
4. In entanglement, what is the relation between the non-local phase and the loop in which the future chooses the past?
5. If the non-local phase is a real fact, what is the main physical consequence that current physics sees but does not explain?

11. Who did what, and where to read

The ontology and the premises are the author's, Ugo Lanciano, who is not a physicist. The translations of the premises into models, the programs and the calculations were made with artificial intelligence (Claude, by Anthropic), in a dialogue directed by the author; a second system (gpt-6-luna, via Codex) checked the calculations. Predictions are registered with a SHA-256 fingerprint and UTC time before the calculation, with the exceptions declared in the deposits; no work has been reviewed by a human referee.

The arithmetic identities used here are not new; what is new is the reading that connects them. Almost every statement of this text refers to a deposit, where the programs and registration logs are; the exception is route V8 on the dimensions (§3), not yet deposited. The single calculation of §6 (version 2.68.1) is in this deposit, in the folder CONTO_UNICO.

What	Deposit	DOI
The book: «Il grande rovesciamento» (The Great Reversal), «Logica e bilancia» (Logic and Scales)	*NUMERI* (2026)	10.5281/zenodo.21802475
General framework and premises	MM 2.30	10.5281/zenodo.22730712
Entry page, with the list of records	MM 2.27	10.5281/zenodo.22667516
Four-page overview of the framework	«The number 1 in front of the	10.5281/zenodo.23535916

	mirror»	
The clock that counts copies	MM 2.40	10.5281/zenodo.23280168
The pairwise rule and the dimensions	MM 2.40.2	10.5281/zenodo.23631239
Born, Tsirelson and the loop; the Born rule from counting copies	MM 2.48 and 2.48.1	10.5281/zenodo.23352814; 10.5281/zenodo.23655646
Deuterium as a present reading	MM 2.58	10.5281/zenodo.23521754
Objects as a common part	MM 2.62.1	10.5281/zenodo.23643717
Mass as a contained multiple; the direction of time	MM 2.64.1	10.5281/zenodo.23643718
The mass threshold of quantum behaviour	MM 2.66	10.5281/zenodo.23643759

What this text adds to the sheet of 11 October. It starts from the author's sheet for a philosopher of 11 October 2026 and updates it with the work up to the 26th:

- what «non-local» means: which of the four senses the framework uses, how the non-local enters the local and what is not claimed (§1, §5);
- the pairwise rule: n counts give one time and n − 1 directions of space (§2, §3);
- objects as the common part of the readings of two observers (§4);
- mass as a contained multiple, invariant across readings (§4);
- the three premises on quantum mechanics: correlated copies, loop, mass threshold (§5);
- the Born rule from counting copies, with three declared assumptions (§5);
- the direction of time as the growth of relations in the observer's matrix (§6);
- Pythagoreanism as the oldest neighbour and the neighbours on non-locality (§8), and the author's answers to the philosophical objections (§10);
- in version 2.68.1, the single calculation that translates the four-level reading and the author's answer on what the future chooses (§6, §10).

How this text was made

The text was drafted in Italian by the AI (Claude) from the author's sheet of 11 October, his premises and the deposits listed above, in a dialogue with the author, who asked for it on 26 October 2026. The external references were checked on 26 October by three separate searches on opened pages (journal, arXiv, Crossref or encyclopaedia); the corrections they found are listed in REFERENCE_CHECK.md. A second instance of the same AI then re-read the whole Italian draft against the deposits and the author's exact words and made 32 remarks; most were applied. Those not applied, or applied only in part, are listed with the reason in REFERENCE_CHECK.md. After an external review pasted by the author, the AI wrote seven explanatory premises. A second system, Gemini, reviewed them with 12 remarks; 11 were applied, and the one not applied proposed a closed question for the author. The author accepted the explanatory premises into the text on 26 October. Later the same day, at the author's request, the AI and Gemini assessed separately how the text treated non-locality; the subsection of §1 on what «non-local» means and its links in §5, §8, §9 and §10 came out of that assessment, and Gemini reviewed them: no serious remarks, five minor ones, all applied, one with narrower words and one in part. A further external review, pasted by the author, led to two more references (Zurek 2005; Schlosshauer and Fine 2005) and to stating which of Bell's assumptions the premise and the loop model each drop. Gemini's reviews, of the explanatory premises, of this package and of the part on non-locality, are in the folder REVIEW_GEMINI. A final check of the whole English text by Gemini found no serious remarks; its four minor ones were applied.

Version 2.68.1 adds the single calculation of §6. Its protocol and criteria were registered before the programs and the calculations; two systems checked it in two rounds each (Astra, gpt-6-luna through Codex, and Gemini), and their remarks, with what was done with each, are in CONTO_UNICO/04_verifica. The author's answer on what the future chooses came after the calculation and is quoted with its date.

References

Each reference was checked on 26 October 2026 on an opened page, given beside it. Quotations from works in other languages are rendered in English.

Philosophy

- Adlam, E., Rovelli, C. (2023), «Information is Physical: Cross-Perspective Links in Relational Quantum Mechanics», *Philosophy of Physics* 1(1): 4. https://philosophyofphysics.lse.ac.uk/articles/10.31389/pop.8
- Aristotle: Huffman, C. (2024), «Pythagoreanism», *Stanford Encyclopedia of Philosophy*, on *Metaphysics* I 5. https://plato.stanford.edu/entries/pythagoreanism/
- Armstrong, D. M. (1983), *What Is a Law of Nature?*, Cambridge University Press. https://plato.stanford.edu/entries/laws-of-nature/
- Beebee, H. (2000), «The Non-Governing Conception of Laws of Nature», *Philosophy and Phenomenological Research* 61(3): 571–594. https://eprints.whiterose.ac.uk/id/eprint/184496/
- Bird, A. (2007), *Nature's Metaphysics: Laws and Properties*, Oxford University Press. https://ndpr.nd.edu/news/nature-s-metaphysics-laws-and-properties/
- Bohm, D. (1980), *Wholeness and the Implicate Order*, Routledge. https://en.wikipedia.org/wiki/Wholeness_and_the_Implicate_Order
- Brouwer, L. E. J. (1981), *Brouwer's Cambridge Lectures on Intuitionism*, ed. D. van Dalen, Cambridge University Press, pp. 4–5. https://plato.stanford.edu/entries/intuitionism/
- Carnap, R. (1950), «Empiricism, Semantics, and Ontology», *Revue Internationale de Philosophie* 4: 20–40. https://plato.stanford.edu/entries/carnap/
- Chakravartty, A. (1998), «Semirealism», *Studies in History and Philosophy of Science A* 29(3): 391–408. https://plato.stanford.edu/entries/structural-realism/
- Einstein, A. (1921), *Geometrie und Erfahrung*, Springer; English translation «Geometry and Experience», in *Sidelights on Relativity*, Methuen, 1922. https://mathshistory.st-andrews.ac.uk/Extras/Einstein_geometry/
- Esfeld, M., Lam, V. (2008), «Moderate structural realism about space-time», *Synthese* 160(1): 27–46. https://iris.unil.ch/lucky-search?index=pid&value=serval:BIB_12BD19C94207
- Franklin, J. (2014), *An Aristotelian Realist Philosophy of Mathematics: Mathematics as the Science of Quantity and Structure*, Palgrave Macmillan. https://ci.nii.ac.jp/ncid/BD08148736
- Howard, D. (1985), «Einstein on Locality and Separability», *Studies in History and Philosophy of Science A* 16(3): 171–201. https://api.crossref.org/works/10.1016/0039-3681(85)90001-9
- Huggett, N., Wüthrich, C. (2025), *Out of Nowhere: The Emergence of Spacetime in Theories of Quantum Gravity*, Oxford University Press. https://academic.oup.com/book/60016
- Kant: Shabel, L. (2021), «Kant's Philosophy of Mathematics», *Stanford Encyclopedia of Philosophy*. https://plato.stanford.edu/entries/kant-mathematics/
- Ladyman, J. (1998), «What Is Structural Realism?», *Studies in History and Philosophy of Science A* 29(3): 409–424; Ladyman, J., Ross, D. (2007), *Every Thing Must Go: Metaphysics Naturalized*, Oxford University Press. https://plato.stanford.edu/entries/structural-realism/
- Lewis, D. (1986), *On the Plurality of Worlds*, Blackwell. https://plato.stanford.edu/entries/lewis-metaphysics/
- Maudlin, T. (2007), *The Metaphysics Within Physics*, Oxford University Press. https://plato.stanford.edu/entries/laws-of-nature/
- Maudlin, T. (2011), *Quantum Non-Locality and Relativity: Metaphysical Intimations of Modern Physics*, 3rd ed., Wiley-Blackwell. https://gov.wiley.com/WileyCDA/WileyTitle/productCd-1444331264,descCd-authorInfo.html
- Newman, M. H. A. (1928), «Mr. Russell's "Causal Theory of Perception"», *Mind* 37(146): 137–148. https://plato.stanford.edu/entries/structural-realism/
- Philolaus: Huffman, C. A. (1993), *Philolaus of Croton: Pythagorean and Presocratic*, Cambridge University Press, fragments 1 and 4. https://plato.stanford.edu/entries/philolaus/
- Psillos, S. (2001), «Is Structural Realism Possible?», *Philosophy of Science* 68(S3): S13–S24. https://plato.stanford.edu/entries/structural-realism/
- Pythagoras: Huffman, C. (2024), «Pythagoras», *Stanford Encyclopedia of Philosophy*. https://plato.stanford.edu/entries/pythagoras/
- Rovelli, C. (1996), «Relational Quantum Mechanics», *International Journal of Theoretical Physics* 35(8): 1637–1678. https://arxiv.org/abs/quant-ph/9609002
- Saunders, S. (2003), «Physics and Leibniz's Principles», in K. Brading, E. Castellani (eds), *Symmetries in Physics: Philosophical Reflections*, Cambridge University Press, pp. 289–308. https://plato.stanford.edu/entries/identity-indiscernible/
- Strawson, P. F. (1959), *Individuals: An Essay in Descriptive Metaphysics*, Methuen. https://plato.stanford.edu/entries/strawson/
- Taylor, E. F., Wheeler, J. A. (1992), *Spacetime Physics*, 2nd ed., W. H. Freeman. https://phys.libretexts.org/Bookshelves/Relativity/Spacetime_Physics_(Taylor_and_Wheeler)/01%3A_Spacetime_Overview/1.01%3A_Parable_of_the_Surveyors
- Tegmark, M. (2008), «The Mathematical Universe», *Foundations of Physics* 38(2): 101–150. https://arxiv.org/abs/0704.0646
- Teller, P. (1986), «Relational Holism and Quantum Mechanics», *British Journal for the Philosophy of Science* 37(1): 71–81. https://api.crossref.org/works/10.1093/bjps/37.1.71
- van Fraassen, B. C. (2006), «Structure: Its Shadow and Substance», *British Journal for the Philosophy of*

Science 57(2): 275–307. https://plato.stanford.edu/entries/structural-realism/
- Vazza, F. (2025), «Astrophysical constraints on the simulation hypothesis for this Universe: why it is (nearly) impossible that we live in a simulation», *Frontiers in Physics* 13: 1561873. https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2025.1561873/full
- Wheeler, J. A. (1990), «Information, Physics, Quantum: The Search for Links», in W. H. Zurek (ed.), *Complexity, Entropy, and the Physics of Information*, Addison-Wesley. https://sfipress.org/books/complexity-entropy-and-the-physics-of-information
- Wigner, E. P. (1960), «The Unreasonable Effectiveness of Mathematics in the Natural Sciences», *Communications on Pure and Applied Mathematics* 13(1): 1–14. https://api.crossref.org/works/10.1002/cpa.3160130102
- Wittgenstein, L. (1922), *Tractatus Logico-Philosophicus*, transl. C. K. Ogden, Kegan Paul, prop. 7. https://en.wikisource.org/wiki/Tractatus_Logico-Philosophicus/7

Physics

- Aharonov, Y., Bergmann, P. G., Lebowitz, J. L. (1964), «Time symmetry in the quantum process of measurement», *Physical Review* 134(6B): B1410–B1416. https://api.crossref.org/works/10.1103/PhysRev.134.B1410
- Argaman, N. (2010), «Bell's theorem and the causal arrow of time», *American Journal of Physics* 78: 1007–1013. https://arxiv.org/abs/0807.2041
- Bell, J. S. (1964), «On the Einstein Podolsky Rosen paradox», *Physics Physique Fizika* 1(3): 195–200. https://api.crossref.org/works/10.1103/PhysicsPhysiqueFizika.1.195
- BIPM (1983), 17th General Conference on Weights and Measures, Resolution 1, «Definition of the metre». https://www.bipm.org/en/committees/cg/cgpm/17-1983/resolution-1
- Bondi, H. (1964), *Relativity and Common Sense*, Doubleday. https://archive.org/details/relativitycommon0000bond_v2s1
- Bonk, M., Schramm, O. (2000), «Embeddings of Gromov hyperbolic spaces», *Geometric and Functional Analysis* 10(2): 266–306. https://api.crossref.org/works/10.1007/s000390050009
- Coll, B., Morales, J. A. (1991), «Symmetric frames on Lorentzian spaces», *Journal of Mathematical Physics* 32(9): 2450–2455.
- Cramer, J. G. (1986), «The transactional interpretation of quantum mechanics», *Reviews of Modern Physics* 58(3): 647–687. https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.58.647
- Derrick, G. H. (1981), «On a completely symmetric choice of space-time coordinates», *Journal of Mathematical Physics* 22(12): 2896–2902.
- Diósi, L. (1987), «A universal master equation for the gravitational violation of quantum mechanics», *Physics Letters A* 120(8): 377–381. https://api.crossref.org/works/10.1016/0375-9601(87)90681-5
- Ehrenfest, P. (1917), «In what way does it become manifest in the fundamental laws of physics that space has three dimensions?», *Proceedings of the Amsterdam Academy* 20 (1918): 200–209. https://link.springer.com/article/10.1007/s00016-004-0221-3
- Gács, P., Körner, J. (1973), «Common information is far less than mutual information», *Problems of Control and Information Theory* 2: 149–162. https://iris.uniroma1.it/handle/11573/16907
- Jennings, D., Rudolph, T. (2010), «Entanglement and the thermodynamic arrow of time», *Physical Review E* 81: 061130. https://arxiv.org/abs/1002.0314
- Knuth, K. H., Bahreyni, N. (2014), «A potential foundation for emergent space-time», *Journal of Mathematical Physics* 55: 112501. https://arxiv.org/abs/1209.0881
- Knuth, K. H. (2016), «The Dimensions of Emergent Spacetime in the Influence Network», APS March Meeting, abstract L44.00014. https://meetings-archive.aps.org/mar/2016/l44/14
- Latune, C. L., Sinayskiy, I., Petruccione, F. (2019), «Heat flow reversals without reversing the arrow of time», *Physical Review Research* 1: 033097. https://journals.aps.org/prresearch/abstract/10.1103/PhysRevResearch.1.033097
- Margolus, N., Levitin, L. B. (1998), «The maximum speed of dynamical evolution», *Physica D* 120: 188–195. https://arxiv.org/abs/quant-ph/9710043
- Micadei, K. et al. (2019), «Reversing the direction of heat flow using quantum correlations», *Nature Communications* 10: 2456. https://arxiv.org/abs/1711.03323
- Panda, C. D. et al. (2024), «Coherence limits in lattice atom interferometry at the one-minute scale», *Nature Physics* 20: 1234–1239. https://arxiv.org/abs/2210.07289
- Pedalino, S. et al. (2026), «Probing quantum mechanics with nanoparticle matter-wave interferometry», *Nature* 649: 866–870. https://api.crossref.org/works/10.1038/s41586-025-09917-9
- Penrose, R. (1996), «On gravity's role in quantum state reduction», *General Relativity and Gravitation* 28(5): 581–600. https://api.crossref.org/works/10.1007/BF02105068
- Planck Collaboration (2020), «Planck 2018 results. VI. Cosmological parameters», *Astronomy & Astrophysics* 641: A6, eq. 47b. https://arxiv.org/abs/1807.06209
- Poh, H. S. et al. (2015), «Approaching Tsirelson's bound in a photon pair experiment», *Physical Review Letters* 115: 180408. https://arxiv.org/abs/1506.01865

- Schlosshauer, M., Fine, A. (2005), «On Zurek's derivation of the Born rule», *Foundations of Physics* 35(2): 197–213. https://arxiv.org/abs/quant-ph/0312058
- Sorkin, R. D. (1994), «Quantum mechanics as quantum measure theory», *Modern Physics Letters A* 9(33): 3119–3127. https://arxiv.org/abs/gr-qc/9401003
- Susskind, L. (1995), «The World as a Hologram», *Journal of Mathematical Physics* 36: 6377–6396. https://arxiv.org/abs/hep-th/9409089
- Swingle, B. (2012), «Entanglement Renormalization and Holography», *Physical Review D* 86: 065007. https://arxiv.org/abs/0905.1317
- Tegmark, M. (1997), «On the dimensionality of spacetime», *Classical and Quantum Gravity* 14: L69–L75. https://arxiv.org/abs/gr-qc/9702052
- 't Hooft, G. (1993), «Dimensional Reduction in Quantum Gravity», arXiv:gr-qc/9310026. https://arxiv.org/abs/gr-qc/9310026
- Touboul, P. et al. (2022), «MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle», *Physical Review Letters* 129: 121102. https://arxiv.org/abs/2209.15487
- Van Raamsdonk, M. (2010), «Building up spacetime with quantum entanglement», *General Relativity and Gravitation* 42: 2323–2329. https://arxiv.org/abs/1005.3035
- Wharton, K. B., Argaman, N. (2020), «Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics», *Reviews of Modern Physics* 92(2): 021002. https://api.crossref.org/works/10.1103/RevModPhys.92.021002
- Wigner, E. (1939), «On unitary representations of the inhomogeneous Lorentz group», *Annals of Mathematics* 40(1): 149–204. https://www.jstor.org/stable/1968551
- Zeeman, E. C. (1964), «Causality implies the Lorentz group», *Journal of Mathematical Physics* 5(4): 490–493. https://api.crossref.org/works/10.1063/1.1704140
- Zurek, W. H. (2005), «Probabilities from Entanglement, Born's Rule from Envariance», *Physical Review A* 71: 052105. https://arxiv.org/abs/quant-ph/0405161
- Zurek, W. H. (2009), «Quantum Darwinism», *Nature Physics* 5: 181–188. https://arxiv.org/abs/0903.5082

Contents of the package

- README.md: this text (English).
- LEGGIMI_IT.md: the same text in Italian.
- AUTHOR_BACKGROUND.md: the author's sentences quoted here, with the Italian originals, dates and context.
- REFERENCE_CHECK.md: how the references were checked, the corrections found, and the review remarks not applied.
- figure_nexus_EN.svg, figura_nesso_IT.svg: Figure 1 in English and Italian.
- REVIEW_GEMINI: the reviews by the second system, Gemini, with what was done with each remark (in Italian).
- CONTO_UNICO: the single calculation of §6 (version 2.68.1): protocol, programs, results, outcome notes and the two rounds of checks, with their registration logs (in Italian; see CONTO_UNICO/README.md).
- LICENSE.md: CC BY 4.0.
- IMPRONTE.sha256: SHA-256 fingerprints of all the files above.

The only programs are those of the single calculation, in CONTO_UNICO; the other calculations cited are in the deposits listed in section 11, with their programs and registration logs.